

Cheng (Kevin) Shao

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EDUCATION

University of Waterloo

B.A.Sc. Electrical Engineering, Honours, Co-op

Waterloo, ON

Sep. 2025 – Expected May 2030

EXPERIENCE

Electrical Team Member

University of Waterloo Baja SAE Team

Jan. 2026 – Present

Waterloo, ON

- Integrating and testing a Data Acquisition (DAQ) system using Arduino, MPU6050 accelerometers, and RPM sensors.
- Researching and coding strain gauges to extract live telemetry data, aiding in comprehensive system calibration.
- Fabricating space-efficient wire harnesses and designing circuit schematics for 2026 competition vehicles.

Project Team Member

Waterloo Experience (WE) Accelerate Program

Winter 2026

Remote

- Conducted accessibility and usability audits of five co-op community pages to identify UI/UX failures.
- Developed digital accessibility micro-course modules focusing on accessible HTML, CSS, and web design.
- Evaluated digital environments for strict AODA compliance and WCAG standard adherence.

Manual Labour Worker

Vanderkooi Farm

Dec. 2024 – Jul. 2025

Nanaimo, BC

- Collaborated with a fast-paced team to execute large-scale livestock transfer operations.
- Processed over 20,000 chickens per shift while strictly adhering to humane handling requirements.
- Maintained high performance and reliability during intense, high-volume seasonal peak periods.

PROJECTS

19V-13V Buck Converter Board (Baja SAE) | *KiCad, LM2596, Hardware Design* Jan. 2026 – Present

- Designed a custom step-down switching regulator PCB to drop a 19V drill battery input to a stable 13V output, reliably powering the vehicle's telemetry electronics.
- Engineered the schematic and routed the board using KiCad, incorporating an LM2596T-ADJ regulator, a 68 μ H (4A Isat) inductor, a 1N5822 Schottky diode, and Low-ESR capacitors for high-efficiency power conversion.

RPM Monitoring Device | *ESP32, C++, HW-477 Hall Effect Sensor*

Jan. 2026 – Present

- Engineered a custom rotational speed sensing solution using an ESP32 and an HW-477 Hall Effect sensor.
- Developed firmware to process analog magnetic signals and calculate accurate, real-time RPM telemetry.
- Integrated the sensor array into a broader vehicle DAQ framework to optimize off-road performance.

Smartwatch Sleep Monitor | *ECE 198 Design Project, Arduino, C++*

Sep. 2025 – Dec. 2025

- Prototyped a wearable Arduino UNO R4 and oximeter sleep tracker to address hospital-induced delirium.
- Programmed firmware to detect sleep cycles via heart rate variability thresholds (Start < 55, End > 65 BPM).
- Implemented IoT functionality to transmit automated daily sleep reports to caregivers via Wi-Fi.

Trades & Mechanical Projects | *VIU Trades Program*

Sep. 2024 – Feb. 2025

- Electrical: Fabricated extension units from technical schematics, ensuring CSA grounding compliance.
- Mechanical: Performed automotive maintenance and brake replacements, prioritizing workshop safety.
- Woodworking & Welding: Fabricated custom products with power tools and fused metals using thermal cutting.

TECHNICAL SKILLS

Programming/Firmware: C++, HTML, Arduino (Wiring)

Hardware & Tools: KiCad, ESP32, Arduino, Accelerometers (MPU6050), Multimeters, Soldering, Power Tools

Relevant Skills: PCB Design, Circuit Assembly, Wire Harnessing, Data Acquisition (DAQ), Technical Documentation

Spoken Languages: English (Fluent), Mandarin (Fluent Speaking/Listening, Intermediate Reading/Writing)